



universidade de aveiro

SEGURANÇA EM REDES DE COMUNICAÇÕES

LAYER 2 ACCESS CONTROL

802.1X & RADIUS

A Linux RADIUS server may be implemented as a **docker** Linux container (based on Linux Debian) with freeradius installed.

With the provided Dockerfile, create a custom docker image:

```
$ docker build -t local/radiusserver -f Dockerfile_radiusserver .
```

Add the Docker container to GNS3 as template:

- Choosing an existing image;
- Define it as local/radiusserver container image;
- Activate 1 network interface (adapter), and
- **Add the following directories as permanent in the advanced tab:**
/etc/freeradius/
/etc/init.d/

A generic Linux PC client (supplicant supporting 802.1X) may be implemented as a **docker** Linux container (based on Linux Alpine) with wpa_supplicant installed.

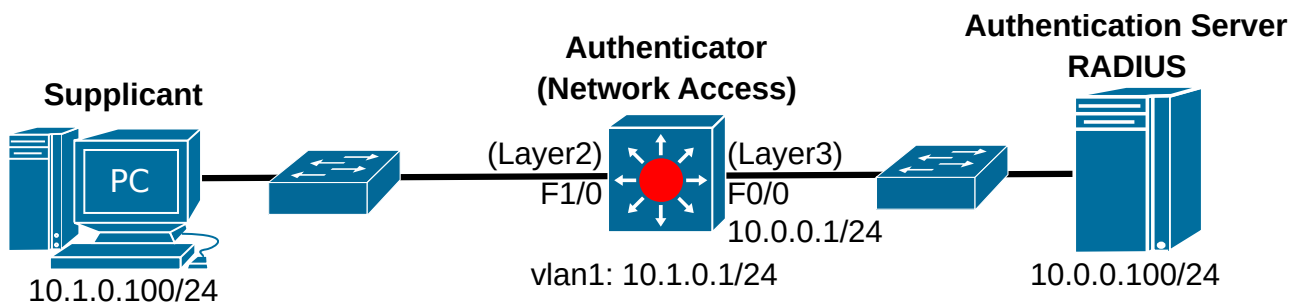
With the provided Dockerfile, create a custom docker image:

```
$ docker build -t local/pcclient -f Dockerfile_pcclient .
```

Add the Docker container to GNS3 as template:

- Choosing an existing image;
- Define it as local/pcclient container image;
- Activate 1 network interface (adapter), and
- **Add the following directories as permanent in the advanced tab:**
/etc/wpa_supplicant/
/etc/init.d/

Note: For both docker images, the Linux bash commands are not permanent (addresses and gateways)! They will disappear upon a reboot. To make them permanent write them in file **/etc/init.d/adv_network**. These commands will be run on every start of the **Docker container**.



1. Set up a network in GNS3 with a client PC (docker local/pcclient), a Layer3 switch, and an authentication RADIUS server (docker local/radiusserver). The client PC will be the an AAA supplicant and the Layer3 switch will be the AAA network authenticator. The PC is connected to a Layer2 access port (VLAN 1) on the Layer3 Switch and the Layer3 Switch connects to the RADIUS server using a Layer3 port.

>> Activate IPv4 routing in the Layer3 Switch with the command: ip routing

>> Configure all IPv4 addresses and gateways.

>> Test the connectivity between all network devices.

2. Configure the RADIUS service: edit the file `/etc/freeradius/3.0/clients.conf` and add the access credentials for the Authenticator/Layer3 switch (10.0.0.1):

```
client SWL3 {
    ipaddr = 10.0.0.1
    secret = radiuskey
}
```

Define the user/supplicant credentials by editing the file `/etc/freeradius/3.0/users` and adding the line
"labredes" Cleartext-Password := "labcom"

Note: confirm that this is not the last line of the configuration file.

Restart and check the status of the radius server, with:

```
RADIUS# service freeradius start
RADIUS# service freeradius status
```

3. Configure the Authenticator by enabling 802.1X in Layer2 ports using the RADIUS server:

```
ESW1(config)# aaa new-model
ESW1(config)# aaa authentication dot1x default group radius
ESW1(config)# dot1x system-auth-control
ESW1(config)# radius-server host 10.0.0.100 auth-port 1812 key radiuskey
ESW1(config)# interface FastEthernet1/0
ESW1(config-if)# dot1x port-control auto
```

In the Layer3 Switch, to verify the status of the 802.1X authentication in port f1/0, run the command:

```
ESW1# show dot1x
```

Start a packet capture between the PC and Layer3 Switch. Test the connectivity between the PC and Layer3 switch.

>> What can you conclude about the PC connectivity?

>> Explain the purpose of the EAP "Request, Identity" packets.

4. Configure the PC/Supplicant 802.1X to use TTLS+PAP, by creating the file `wired.conf`, in directory `/etc/wpa_supplicant`, with the following contents:

```
ctrl_interface=/run/wpa_supplicant
ap_scan=0
```

```
network={
    key_mgmt=IEEE8021X
    eap=TTLS
    identity="labredes"
    password="labcom"
    phase2="auth=PAP"
}
```

Start a packet capture between the PC and Layer3 Switch (filter `eapol`), and another between the Layer3 Switch and the RADIUS server (filter `radius`).

Run the following command in bash to initiate the Ethernet/wired 802.1X negotiation with the Authenticator (in debug mode).

```
PC# wpa_supplicant -D wired -i eth0 -c /etc/wpa_supplicant/wired.conf -dd
```

In the Layer3 Switch, to verify the status of the 802.1X authentication in port f1/0, run the command:

```
ESW1# show dot1x
```

>> What can you conclude about the result of the 802.1X authentication process?

>> Test the connectivity between the PC and the Layer3 switch (and network 10.0.0.0/24).

>> Analyze and explain the purpose of the EAP and TLSv1 packets exchanged between the PC and Layer3 switch.

>> Analyze and explain the purpose and content of the RADIUS packets exchanged between the Layer3 switch and the RADIUS server.

5. Activate the RADIUS log functionality to include the logging of authentication requests. Edit the file /etc/freeradius/3.0/radiusd.conf and change the following value inside the log section:

auth = yes

Restart the RADIUS server: service freeradius restart

On the PC, test multiple accesses with correct and wrong usernames and passwords.

>> Analyze the latest RADIUS log file in /var/log/freeradius and identify the successful and unsuccessful authentication (Auth) entries.

>> How, in a centralized manner, detect failed access attempts and identifying the terminal MAC address and location (switch/port) were it occurred?

Note: if the authentication process freezes on the switch, perform a shutdown/no shutdown on that port.

6. On your own PC, identify where to configure 802.1X using the GUI.